

Verifying Citations Against Open Scholarly Databases

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Abstract

Fabricated citations are entering the published literature at a rate that has climbed roughly tenfold in three years. We describe a verification pipeline that resolves extracted references against Crossref and OpenAlex, checks retraction status against the Retraction Watch database, and reports resolved, unresolved, retracted, or metadata mismatch for every reference in a paper.

1 Introduction

A citation can fail in two distinct ways: it can be fabricated, or it can be real but misrepresented. Existence checking catches the first failure. It cannot catch the second, because a bibliographically valid citation that does not support the claim it is attached to passes every existence check.

2 Verification Pipeline

Every extracted reference is resolved against Crossref by DOI when available, falling back to a fuzzy title and author match against OpenAlex. Retraction status is checked against the Retraction Watch database, which Crossref has published openly since 2023.

3 Results

Across a sample of extracted reference lists, the resolver correctly identified retracted references that existence-only checking would have passed as valid, and flagged a small number of metadata mismatches between the citing paper's record and the resolved source.

4 Conclusion

Verification against open scholarly infrastructure is inexpensive and catches a failure mode that peer review does not systematically check today.

References

- Statnews (2026). Lancet study finds steep rise in fraudulent citations. statnews.com.
- Crossref (2023). Retraction Watch data now in the Crossref API. crossref.org/blog.
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